

Solutions for Homework Assignment #4

Answer to Question 1.

• **A IS UNDECIDABLE BUT RECOGNIZABLE AND $\bar{A}$ IS UNRECOGNIZABLE.**

JUSTIFICATION:

1. A is undecidable: It suffices to prove that $\text{UNIV} \leq_m A$. To this end we show how, given $\langle M, x \rangle$, we can construct $\langle M_1, M_2 \rangle$ such that:

- if M accepts x then there is a string y such that one of M_1 and M_2 accepts both y and yy ; and
- if M does not accept x then there is no y such that one of M_1 and M_2 accepts both y and yy .

M_1 and M_2 are defined as follows.

$M_1 :=$ on input y : 1 if $y = 0$ then accept 2 else reject	$M_2 :=$ on input y : 1 run M on x (use the universal TM) 2 if M accepts then accept 3 else reject
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With M_1 and M_2 defined this way, we have:

$$\mathcal{L}(M_1) = \{0\} \quad \text{and} \quad \mathcal{L}(M_2) = \begin{cases} \Sigma^* & \text{if } M \text{ accepts } x \\ \emptyset & \text{if } M \text{ does not accept } x \end{cases}$$

Therefore:

- If M accepts x , then for some y (in fact, for all y) one of M_1 and M_2 (in particular, M_2) accepts both y and yy , as wanted.
- If M does not accept x , then for every y neither M_1 nor M_2 accept both y and yy (M_1 accepts only the string 0 and M_2 accepts no string at all), as wanted.

Clearly, the mapping $\langle M, x \rangle \mapsto \langle M_1, M_2 \rangle$ is computable. So $\text{UNIV} \leq_m A$, as claimed.

2. A is recognizable: Following is a recognizer for A , where x_i is the i -th string in shortlex order.

A-RECOGNIZER := on input $\langle M_1, M_2 \rangle$:

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1  for each pair of natural numbers  $(i, j)$  in dovetail order do
2    run  $M_1$  on  $x_i$  for  $j$  moves or until it halts, whichever happens first (use the universal TM)
3    if  $M_1$  accepts then
4      run  $M_1$  on  $x_i x_i$  for  $j$  moves or until it halts, whichever happens first (use the universal TM)
5      if  $M_1$  accepts then accept
6    run  $M_2$  on  $x_i$  for  $j$  moves or until it halts, whichever happens first (use the universal TM)
7    if  $M_2$  accepts then
8      run  $M_2$  on  $x_i x_i$  for  $j$  moves or until it halts, whichever happens first (use the universal TM)
9      if  $M_2$  accepts then accept

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This accepts $\langle M_1, M_2 \rangle$ if and only if there is a string y so that one of M_1 and M_2 accepts both y and yy .

ONLY IF: Suppose there is a string y so that one of M_1 and M_2 accepts both y and yy . Let (i, j) be the first pair (in dovetail order) so that both x_i and $x_i x_i$ are accepted by one of M_1 and M_2 within j moves. Then when the for loop is executed for pair (i, j) , the input $\langle M_1, M_2 \rangle$ will be accepted.

IF: Suppose the above procedure accepts $\langle M_1, M_2 \rangle$ when the for loop is dovetailing through the pair (i, j) . Then the string x_i has the required property.

Thus, the above procedure is a recognizer for A . (It is not a decider because if $\langle M_1, M_2 \rangle \notin A$, it will loop.)

3. $\bar{A}$ is unrecognizable: Suppose, for contradiction, that $\bar{A}$ is recognizable. Since A is also recognizable (proven in (2) above), A would be decidable, contrary to what we proved in (1). Thus, $\bar{A}$ is unrecognizable.

• B IS UNDECIDABLE BUT RECOGNIZABLE AND $\overline{B}$ IS UNRECOGNIZABLE.

JUSTIFICATION:

1. B is undecidable: It suffices to prove that $\text{UNIV} \leq_m B$. To this end we show how, given $\langle M, x \rangle$, we can algorithmically construct $\langle M', y, q \rangle$ such that:

- if M accepts x then M' on input y is in state q in at least two moves; and
- if M does not accept x then M' on input y is not in state q in at least two moves.

M' is a TM that works like M except that it has an additional state q^* and its transition function is modified so that instead of entering the accept state, it enters state q^* and remains in that state thereafter. In more detail, if $M = (Q, \Sigma, \Gamma, \delta, q_0, h_A, h_R)$ then $M' = (Q \cup \{q^*\}, \Sigma, \Gamma, \delta', q_0, h_A, h_R)$, where $q^* \notin Q$ and δ' is defined as follows. For any $q \in Q \cup \{q^*\}$, $p \in Q$, $a, b \in \Gamma$, and $D \in (L, R)$,

$$\delta'(q, a) = \begin{cases} (p, b, D), & \text{if } q \neq q^*, p \neq h_A, \text{ and } \delta(q, a) = (p, b, D) \\ (q^*, -, -), & \text{if } q = q^* \text{ or } p = h_A \text{ (where } - \text{ means "anything") } \end{cases}$$

By construction of M' we have

- If M accepts x then M' is in state q^* in at least two moves (in fact, infinitely many), as wanted.
- If M does not accept x then M' is not in state q^* in at least twice (in fact, never), as wanted.

Clearly, the mapping $\langle M, x \rangle \mapsto \langle M', x, q \rangle$ is computable. Therefore $\text{UNIV} \leq_m B$, as claimed.

2. B is recognizable: Following is a recognizer for B . It simply simulates the given TM M on the given input x and uses a counter to keep track of the number of moves in which M is in state q . If such a counter ever becomes 2, the recognizer accepts; otherwise it keeps simulating M on x until it halts (in which case it rejects) or it keeps going forever (if M never halts and is not in state q in at least two moves).

B -RECOGNIZER := **on input** $\langle M, x, q \rangle$:

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1   if  $q$  is the initial state of  $M$  then  $\text{Ctr} := 1$  else  $\text{Ctr} := 0$ 
2   while true do
3       if  $M$  is in the accept or reject state then reject
4       simulate one more step of  $M$  on input  $x$  (use the universal TM)
5       if  $M$  is in state  $q$  then
6            $\text{Ctr} := \text{Ctr} + 1$ 
7       if  $\text{Ctr} = 2$  then accept
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It is obvious that B -RECOGNIZER accepts $\langle M, x, q \rangle$ if and only if M is in state q in at least two moves, so it is indeed a recognizer for B . (It is not a decider because it loops if M loops on x without being in state q in two moves.)

3. $\overline{B}$ is unrecognizable: Suppose, for contradiction, that $\overline{B}$ is recognizable. Since B is also recognizable (proven in (2) above), B would be decidable, contrary to what we proved in (1). Thus, $\overline{B}$ is unrecognizable.

• C AND $\overline{C}$ ARE BOTH DECIDABLE.

JUSTIFICATION: We can design a decider for C by modifying the recognizer for B given above. We now use different counters, one for each state q of M , to keep track of the number of moves in which M on input x is in state q . By the pigeonhole principle, after at most $k - 1$ moves, where k is the number of states of M (and therefore $k - 2$ is the number of non-halting states of M), if M has not already halted, it will have been in the same state for at least two moves. Thus, in this case we can be sure that the while loop will terminate and the algorithm will always either accept or reject its input. The modified algorithm is shown below.

C -DECIDER := **on input** $\langle M, x \rangle$:

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1   for each state  $q \neq q_0$  of  $M$  do  $Ctr[q] := 0$ 
2    $Ctr[q_0] := 1$ 
3   while true do
4       if  $M$  is in the accept or reject state then reject
5       simulate one more step of  $M$  on input  $x$  (use the universal TM)
6        $q :=$  current state of  $M$ 
7        $Ctr[q] := Ctr[q] + 1$ 
8       if  $Ctr[q] = 2$  then accept

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This algorithm accepts $\langle M, x \rangle$ if and only if M on input x is in the same state in at least two moves. Furthermore, the while loop will terminate after a finite number of iterations for the reason explained above, and so the algorithm will reject $\langle M, x \rangle$ if and only if M on input x is not in the same state in at least two moves. This proves that C is decidable, and therefore so is $\bar{C}$.

• D AND $\bar{D}$ ARE BOTH UNRECOGNIZABLE.

JUSTIFICATION: To prove this it suffices to show that $\text{UNIV} \leq_m D$ and $\text{UNIV} \leq_m \bar{D}$. (Because then, by Theorem 4.6, $\overline{\text{UNIV}} \leq_m \bar{D}$ and $\overline{\text{UNIV}} \leq_m D$. Since $\overline{\text{UNIV}}$ is unrecognizable (Theorem 3.7), by Theorem 4.5, D and $\bar{D}$ are unrecognizable.)

1. $\text{UNIV} \leq_m D$: For this we show how, given $\langle M, x \rangle$, we can algorithmically construct $\langle M' \rangle$ such that:

- if M accepts x then there are exactly five strings accepted by both M_1 and M_2 ; and
- if M does not accept x then the number of strings accepted by both M_1 and M_2 is not five.

M_1 and M_2 are the TMs defined below:

$M_1 :=$ on input y : 1 if $y \in \{\epsilon, 0, 1, 00, 01\}$ then 1 accept 2 else reject	$M_2 :=$ on input y : 1 run M on x (use the universal TM) 2 if M accepts then accept 3 else reject
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With M_1 and M_2 defined this way, we have:

$$\mathcal{L}(M_1) = \{\epsilon, 0, 1, 00, 01\} \quad \text{and} \quad \mathcal{L}(M_2) = \begin{cases} \Sigma^*, & \text{if } M \text{ accepts } x \\ \emptyset, & \text{if } M \text{ does not accept } x \end{cases}$$

Therefore:

- If M accepts x then there are exactly five strings accepted by both M_1 and M_2 (namely the five accepted by M_1), as wanted.
- If M does not accept x then the number of strings accepted by both M_1 and M_2 is not five (it is zero, because M_2 accepts no string), as wanted.

Clearly, the mapping $\langle M, x \rangle \mapsto \langle M_1, M_2 \rangle$ is computable. This shows that $\text{UNIV} \leq_m D$, and therefore $\bar{D}$ is unrecognizable.

2. $\text{UNIV} \leq_m \bar{D}$: For this we show how, given $\langle M, x \rangle$, we can algorithmically construct $\langle M_1, M_2 \rangle$ such that:

- if M accepts x then the number of strings accepted by both M_1 and M_2 is not five; and
- if M does not accept x then there are exactly five strings accepted by both M_1 and M_2 .

M_1 and M_2 are the TMs defined below:

$M_1 :=$ on input y : 1 accept	$M_2 :=$ on input y : 1 if $y \in \{\epsilon, 0, 1, 00, 01\}$ then accept 2 run M on x (use the universal TM) 3 if M accepts then accept 4 else reject
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With M_1 and M_2 defined this way, we have:

$$\mathcal{L}(M_1) = \Sigma^* \quad \text{and} \quad \mathcal{L}(M_2) = \begin{cases} \Sigma^*, & \text{if } M \text{ accepts } x \\ \{\epsilon, 0, 1, 00, 01\}, & \text{if } M \text{ does not accept } x \end{cases}$$

Therefore:

- If M accepts x then the number of strings accepted by both M_1 and M_2 is not five (there are infinitely many such strings, because M_1 and M_2 accept every string), as wanted.
- If M does not accept x then there are exactly five strings accepted by both M_1 and M_2 (namely the strings ϵ , 0, 1, 00, and 01), as wanted.

Clearly, the mapping $\langle M, x \rangle \mapsto \langle M_1, M_2 \rangle$ is computable. This shows that $\text{UNIV} \leq_m \overline{D}$, and therefore D is unrecognizable.

• **E AND $\overline{E}$ ARE BOTH UNRECOGNIZABLE.**

JUSTIFICATION: As we saw for D , it suffices to prove that $\text{UNIV} \leq_m E$ and $\text{UNIV} \leq_m \overline{E}$.

1. $\text{UNIV} \leq_m E$: For this we show how, given $\langle M, x \rangle$, we can algorithmically construct $\langle M' \rangle$ such that:

- If M accepts x then M' accepts every string.
- If M does not accept x then M' does not accept every string.

M' is defined as follows.

$M' :=$ **on input** y :

- 1 run M on x (use the universal TM)
- 2 **if** M accepts x **then accept**
- 3 **else reject**

Therefore,

- If M accepts x , then M' accepts every string, as wanted.
- If M does not accept x , then M' does not accept every string (since it accepts no string at all!), as wanted.

It is clear that the mapping $\langle M, x \rangle \mapsto \langle M' \rangle$ is computable. Thus, $\text{UNIV} \leq_m E$, as wanted.

2. $\text{UNIV} \leq_m \overline{E}$: For this we show how, given $\langle M, x \rangle$, we can algorithmically construct $\langle M' \rangle$ such that:

- if M accepts x then M' does not accept every string; and
- if M does not accept x then M' accepts every string.

M' is defined as follows, using an idea we saw in class when proving the unrecognizability of the set of codes of TMs that recognize finite languages.

$M' :=$ **on input** y :

- 1 run M on x for $|y|$ steps or until it halts, whichever comes first (use the universal TM)
- 2 **if** M accepts x **then reject** (or loop)
- 3 **else accept**

With this construction, we have:

- If M accepts x in, say, t moves, then M' does not accept any string y that has length t or more. So in this case, M' does not accept all strings, as wanted.
- If M does not accept x , then for any string y , M does not accept x in at most $|y|$ steps, and therefore M' accepts y in line 3. So in this case, M' accepts all strings, as wanted.

It is clear that the mapping $\langle M, x \rangle \mapsto \langle M' \rangle$ is computable. Thus, $\text{UNIV} \leq_m \overline{E}$, as wanted.

Answer to Question 2.

a. To prove that $\text{ALL} \equiv_m \text{INFIN}$, it suffices to show that (i) $\text{ALL} \leq_m \text{INFIN}$, and (ii) $\text{INFIN} \leq_m \text{ALL}$.

(i) $\text{ALL} \leq_m \text{INFIN}$. Given $\langle M \rangle$ we want to construct $\langle M' \rangle$ such that

- if M accepts all strings then M' accepts an infinite number of strings; and
- if M does not accept all strings then M' accepts a finite number of strings.

We define such an M' as follows:

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 $M' :=$  on input  $x$ :
1   for each string  $y$  that precedes  $x$  in shortlex order do
2       run  $M$  on  $y$  (use the universal TM)
3       if  $M$  rejects then reject (or loop)
4   accept

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We have:

- If M accepts every string then, for any string x , it accepts every string that precedes x in shortlex order. Therefore, M' on any input x will complete the for loop in line 1 without being stuck in line 2 or rejecting in line 3, and therefore M' accepts x in line 4. So, in this case M' accepts an infinite number of strings (in fact, all of them), as wanted.
- If M does not accept all strings, then there is some first string (in shortlex order), say y^* , that it does not accept — i.e., M either rejects y^* or it loops on y . Then, for every string x that is after y^* in shortlex order, M' will either loop in line 2 in the iteration of the for loop where $y = y^*$ or it will reject in line 3. So, in this case, M' accepts a finite number of strings (those up to y^* in shortlex order), as wanted.

Clearly the mapping $\langle M \rangle \mapsto \langle M' \rangle$ can be constructed by a TM, so $\text{ALL} \leq_m \text{INFIN}$.

(ii) $\text{INFIN} \leq_m \text{ALL}$. Given $\langle M \rangle$ we want to construct $\langle M' \rangle$ such that

- if M accepts an infinite number of strings then M' accepts all strings; and
- if M accepts a finite number of strings then M' does not accept some string.

We define such an M' below. Informally, given any input x , M' seeks a string y that follows x (in shortlex order) that M accepts, and if it finds such a y , M' accepts x . To avoid getting stuck on seeing whether M accepts a string y without ever getting the chance to consider strings that follow y in shortlex order, M' performs this search via dovetailing to limit the number of steps on which it runs M on any input. In this manner, M' gets stuck looking for a string y that follows x and M accepts if and only if no such string exists. We now describe M' more precisely, where x_i denotes the i -th string in shortlex order:

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 $M' :=$  on input  $x$ :
1   for each pair of natural numbers  $(i, j)$  in dovetail order do
2       if  $x_i$  follows  $x$  in shortlex order then
3           run  $M$  on  $x_i$  for  $j$  steps or until it halts, whichever comes first (use the universal Turing machine)
4       if  $M$  accepts then accept

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We have:

- If M accepts an infinite number of strings then, for any string x , M accepts some string y that follows x in shortlex order; let x_i be such a string and let j be the number of steps in which M accepts x_i . Then, in the iteration of the for loop in line 1 that considers the pair (i, j) , M' will accept x — if it hasn't done so already. So, if M accepts an infinite number of strings, M' accepts all strings, as wanted.
- If M accepts a finite number of strings, let x^* be the last string (in shortlex order) that it accepts (or ϵ if M accepts no string). Then M' on input any string y that is after x^* in shortlex order will be stuck forever in the for loop in line 1, as there is no pair (i, j) so that string x_i follows x^* in shortlex order and M accepts x_i in at most j steps, so the conditional in line 4 will always fail. So, in this case, M' does not accept all strings (in particular it does not accept any string that follows x^* in shortlex order), as wanted.

Clearly the mapping $\langle M \rangle \mapsto \langle M' \rangle$ is computable, so $\text{INFIN} \leq_m \text{ALL}$.

b. It suffices to show that $\text{ALL} \not\leq_m \text{UNIV}$. Suppose, for contradiction, that $\text{ALL} \leq_m \text{UNIV}$. Then, since UNIV is recognizable, it follows that ALL is also recognizable. But we have already shown (see set E in Question 1) that ALL is not recognizable. Thus, $\text{ALL} \not\leq_m \text{UNIV}$.